

DaSEH Day 6 Cheatsheet

Data Manipulation

Major concepts

- **Wide data:** Multiple columns per individual, with values spread across columns
- **Long data:** Multiple rows per observation, with values stored in a single column

Functions

Library/Package	Piece of code	Example of usage	What it does
tidyverse (tidyr)	<code>separate()</code>	<code>df > separate(x, c("A", "B"))</code>	Separates a character column into multiple columns using a regular expression or numeric positions
tidyverse (tidyr)	<code>unite()</code>	<code>df > unite("z", x:y, remove = FALSE)</code>	Combines multiple columns into one column
tidyverse (tidyr)	<code>pivot_longer()</code>	<code>df > pivot_longer(!col_to_keep, names_to = "new_col_with_labels", values_to = "new_col_with_values")</code>	Lengthens a data frame by increasing rows and decreasing columns
tidyverse (tidyr)	<code>pivot_wider()</code>	<code>df > pivot_wider(names_from = "col_with_names", values_from = "col_with_values")</code>	Widens a data frame by decreasing rows and increasing columns

Library/Package	Piece of code	Example of usage	What it does
tidyverse (dplyr)	<code>?_join()</code>	<code>inner_join(x, y)</code>	Joins data from two data frames. <code>inner_join(x, y)</code> Keeps only rows that match in both <code>x</code> and <code>y</code>. <code>full_join(x, y)</code> Keeps all rows from <code>x</code> and <code>y</code>. <code>left_join(x, y)</code> Keeps all rows from <code>x</code>, even if they do not match <code>y</code>. <code>right_join(x, y)</code> Keeps all rows from <code>y</code>, even if they do not match <code>x</code>. <code>anti_join(x, y)</code> Keeps rows from <code>x</code> that do not match <code>y</code>, retaining only columns from <code>x</code>.

Data Visualization with **esquisse**

Major concepts

- **esquisse**: Package for testing and learning the syntax for making plots

Functions

Library/Package	Piece of code	Example of usage	What it does
esquisse	<code>esquisser()</code>	<code>esquisser(Orange)</code> <code>esquisser(Orange, viewer = "browser")</code>	Starts an interactive esquisse session to create a plot from a <code>data.frame</code> or <code>tibble</code>

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.